



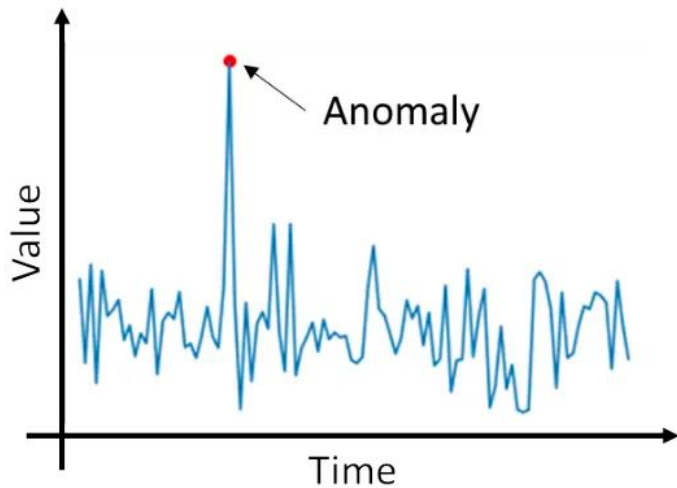
Abdul-Kazeem Shamba

- PhD Candidate at IDI and DATA WP(September 2023)
 - Representation Learning in Sequential Data
- Education:
 - MSc. Marine Intelligent Robotics (NTNU and UTLN) – 2023
 - MSc. Applied Machine Learning (Carnegie Mellon University) – 2021
 - BSc. Electronic and Electrical Eng. (Obafemi Awolowo University) – 2018
- Experience:
 - Computer Vision Engineer at Pure Lobster, Norway (2023)
 - Computer Vision Engineer at Alteia, France(2022)
 - Software Developer at TechnoServe, Benin Rep. (2021)
 - Teaching Assistant at CMU-Africa, Rwanda (2021)
 - Control Engineer at Transmission Company of Nigeria (2019)

Presentation overview

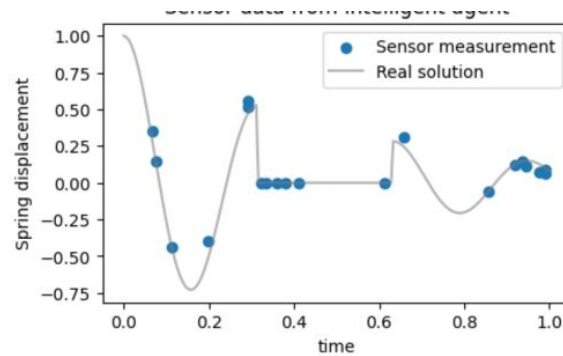
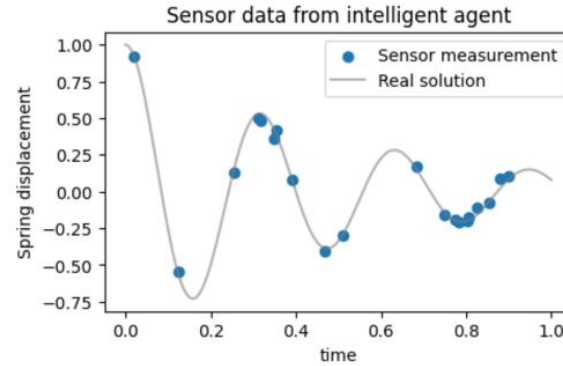
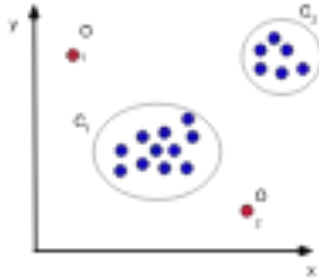
Anomaly Detection (AD):	Types of anomalies, AI techniques for anomaly detection
Hybrid AI:	Explain hybrid AI focusing physics informed neural network
Hybrid AI for AD:	Show result from the DT8124 course project on a toy example
Current Research:	Present current research and future work direction

Anomaly detection



Anomaly detection

•



Anomaly detection

Input

- Temporal or Sequential data
- Spatial data
- Graph data
- Spatio-temporal data

Types

- Point anomaly
- Contextual
- Collective Anomaly

Why detect anomalies

- Indicate Errors
- Significant Events
- Potential Discoveries

Anomaly detection: Techniques

38:6

G. Pang et al.

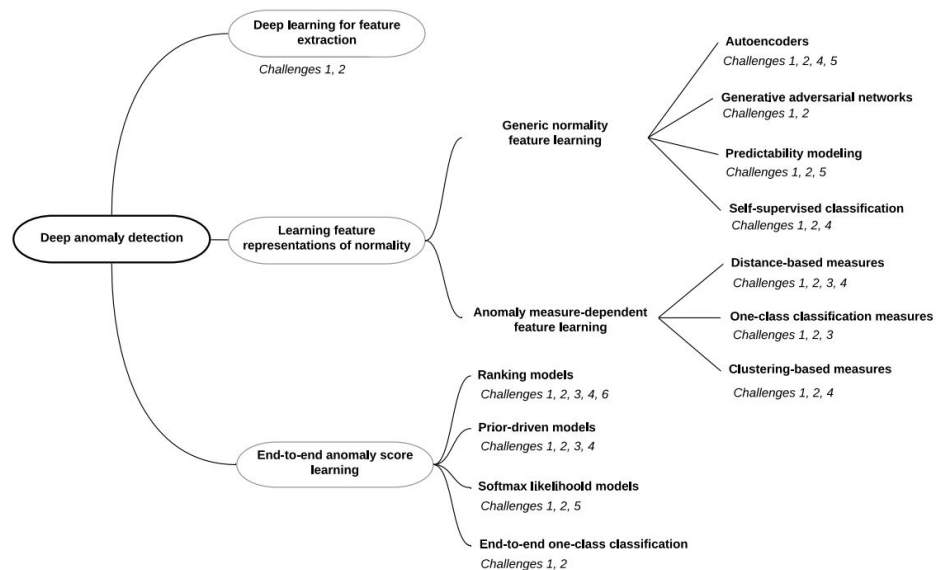


Fig. 1. The proposed taxonomy of current deep anomaly detection techniques. The detection challenges that each category of methods can address are also presented.

Hybrid AI

Hybrid AI

- Symbolic-Connectionist
- Evolutionary Algorithms with ML
- Genetic Programming with ML
- Physics with ML

Physics

- Physics-Guided
- Physics-Informed

2018 ~30 citations
 2019 ~ 100 citations
 2020 ~600 citations
 2024 ~ 7629 citations



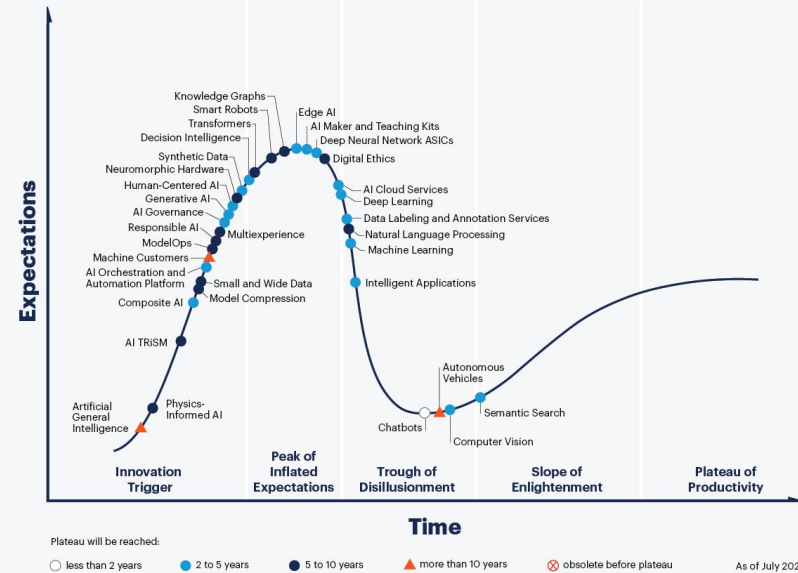
Journal of Computational Physics
 Volume 378, 1 February 2019, Pages 686-707



Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations

M. Raissi^a, P. Perdikaris^b, G.E. Karniadakis^a

Hype Cycle for Artificial Intelligence, 2021



gartner.com

Source: Gartner
 © 2021 Gartner, Inc. and/or its affiliates. All rights reserved. Gartner and Hype Cycle are registered trademarks of Gartner, Inc. and its affiliates in the U.S. 1482644

Gartner.

Hybrid AI for Anomaly Detection



$$m \frac{d^2 u}{dt^2} + \mu \frac{du}{dt} + ku = 0 ,$$

where m is the mass of the oscillator, μ is the coefficient of friction and k is the spring constant and u is the spring displacement.

$$\mathcal{L}(\theta, \mu) = \frac{1}{N} \sum_i^N \left(\left[m \frac{d^2}{dt^2} + \mu \frac{d}{dt} + k \right] u_{\text{PINN}}(t_i; \theta) \right)^2 + \frac{\lambda}{M} \sum_j^M (u_{\text{PINN}}(t_j; \theta) - u_{\text{obs}}(t_j))^2$$

Hybrid AI for Imputation and Anomaly Detection on Sensor Data

*DT8124 - Intelligent Agents: Reasoning, Planning, **Perception** and Cooperation

Abdul-Kazeem Shamba
Department of Computer Science
Norwegian University of Science and Technology
Trondheim, Norway
abdul.k.shamba@ntnu.no

Abstract—This study introduces Physics-Informed Neural Networks (PINNs) as a groundbreaking fusion of data-driven machine learning and physics-based modeling, focusing on sensor data imputation and anomaly detection for intelligent agents. PINNs, enhanced with physics-informed constraints, demonstrate competitive performance in predicting distributions and reconstructing missing sensor readings, particularly in scenarios with sparse data or sensor occlusion. An innovative anomaly detection approach, leveraging a learnable parameter μ , highlights PINNs' robustness compared to traditional neural networks, providing interpretability for real-world applications with subtle yet impactful anomalies. This hybrid AI approach holds significant promise for advancing intelligent agent capabilities, offering insights into the evolving landscape of AI applications in sensor data processing, imputation, and anomaly detection across diverse domains.

Index Terms—Hybrid AI, Imputation, Intelligent Agent, Anomaly Detection, Sensor Data, Perception, Harmonic Oscillator, Physics Informed Neural Networks, Machine Learning

I. INTRODUCTION

Intelligent Agents have seen increased adoption in industries in the last decade [9], [10]. A crucial factor leading to this popularity is the ability of an agent to perceive its environment through sensors and intelligently act upon this data. Intelligent agents gather vast information about their surroundings

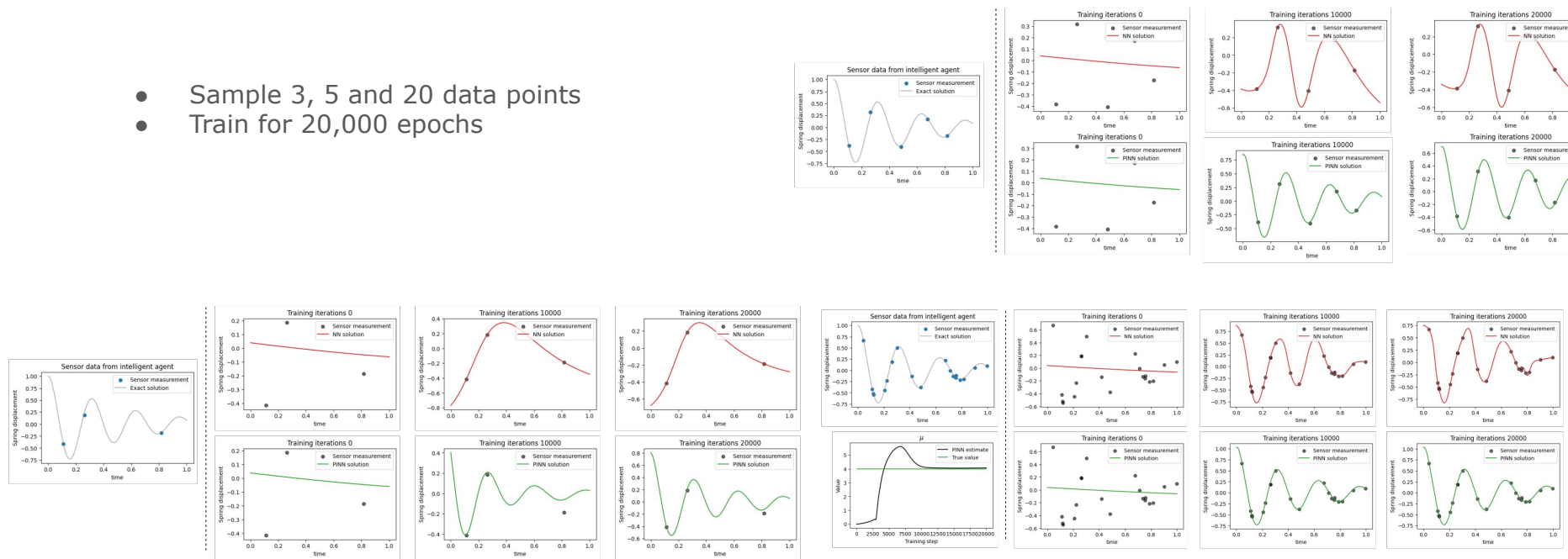
research, we focus on the latter. We acknowledge that two potential problems arise when an agent sends sensor data to a central location. Due to the large storage requirement and the need for compressed sensing, the sensed data are sparse. Therefore, we need an algorithm to perform imputation for the missing sensor data when required.

Another potential problem is occlusion or sensor failure. An anomaly detection algorithm is necessary to flag sensor occlusion and failures. The improved performance of an imputation and anomaly detection algorithm on sensor data from an agent is relevant in enhancing data fidelity and making an agent more intelligent. Classical machine learning (ML) approaches have attempted to solve the imputation and anomaly detection problem in time series data. However, most of these models are prone to overfitting the observed data and do not generalize well on unseen data. Recently, the success of deep neural networks, other natural language processing, and computer vision has spiraled increased application in time series for anomaly detection. Deep neural networks are usually complex with large training parameters, and not interpretable.

Hybrid AI for Anomaly Detection

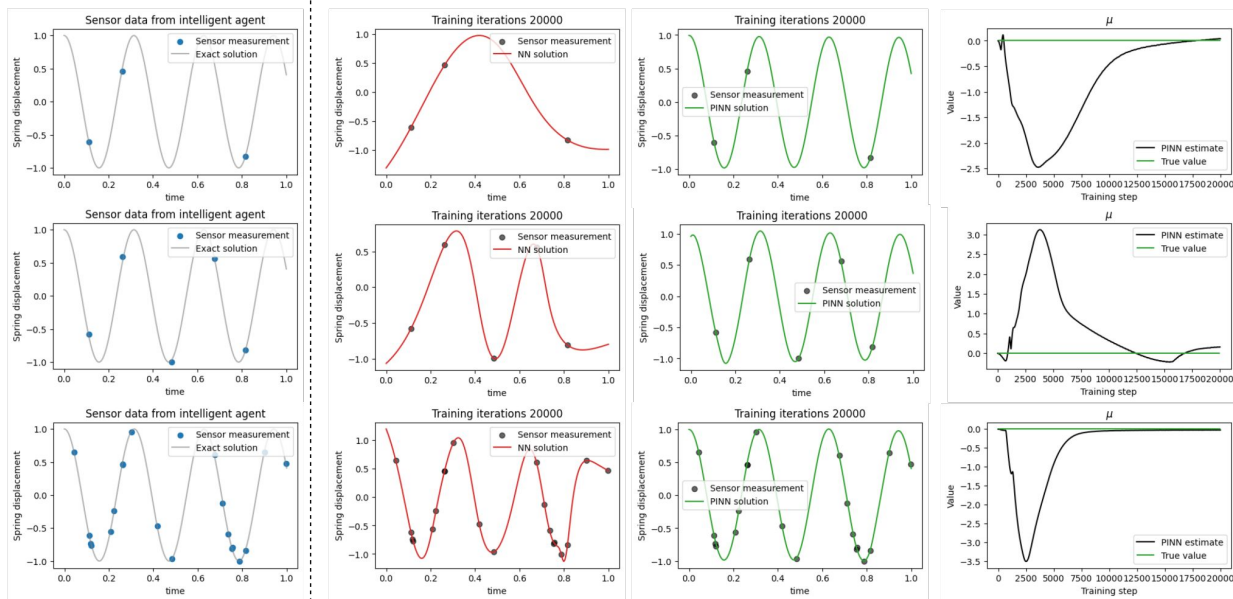
Imputation - Sensor Data Distribution (Underdamped)

- Sample 3, 5 and 20 data points
- Train for 20,000 epochs



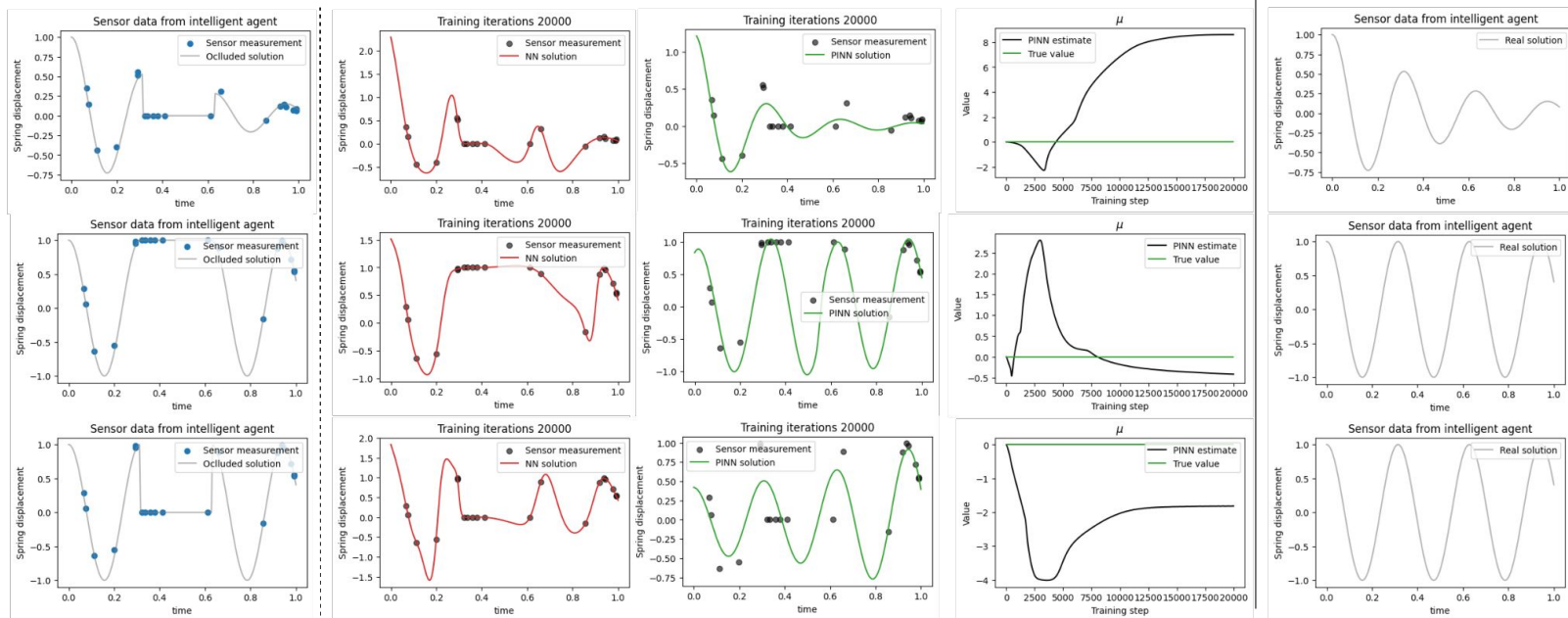
Hybrid AI for Anomaly Detection

Imputation – Sensor Data Distribution (Undamped)



Hybrid AI for Anomaly Detection

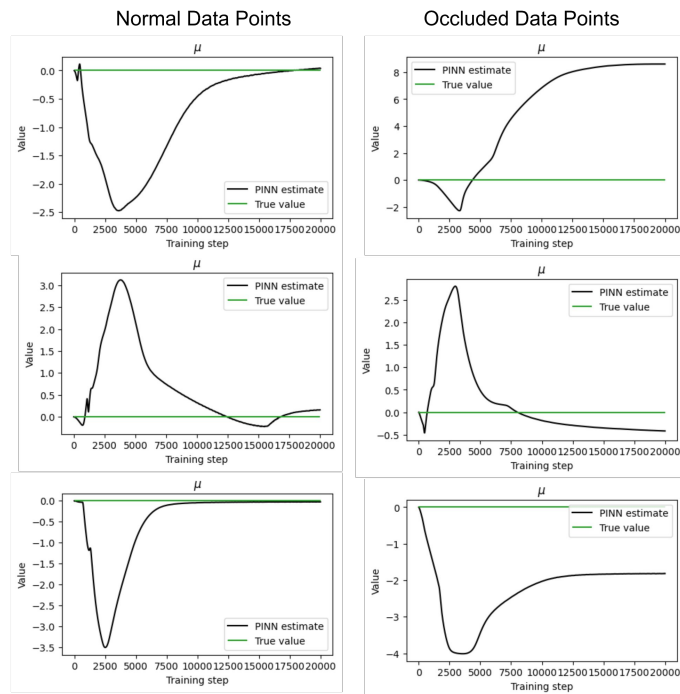
Imputation – Sensor Data Distribution



Hybrid AI for Anomaly Detection

Anomaly Detection - Learning Sensor Data Parameter

- Known learnable parameter
- Create a **threshold**
- Compare value of observed point vs the known (**normal**)
- Flag off an anomaly if this value deviates significantly from the **threshold**



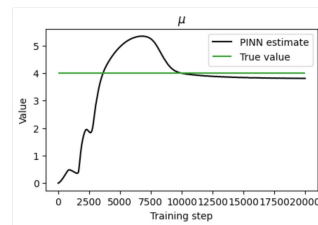
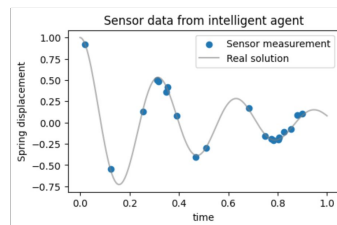
Hybrid AI for Anomaly Detection

Anomaly Detection - Learning Sensor Data Parameter

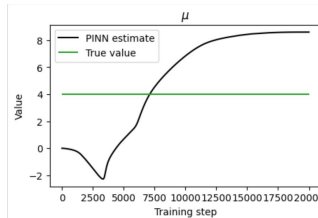
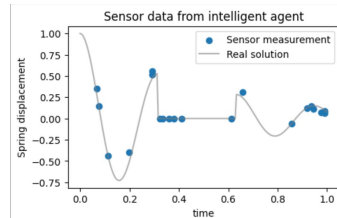
RECONSTRUCTION LOSS AND COEFFICIENT OF FRICTION μ FOR NORMAL AND ANOMALOUS MEASUREMENTS

Anomaly detection parameters		
	MSE loss (NN)	Parameter μ (PINN)
Normal	0.0647	3.8062
Anomaly 1	0.0632	8.6019
Anomaly 2	0.3776	2.4382

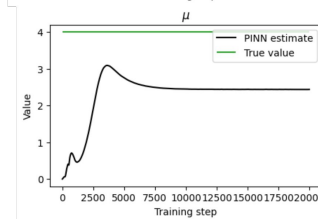
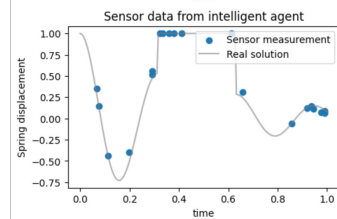
Normal



Anomaly 1

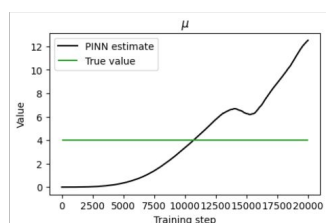
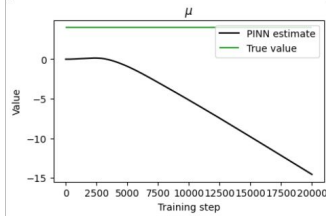
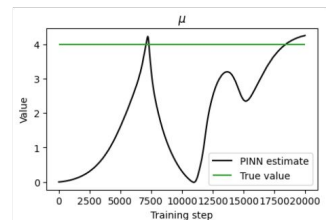
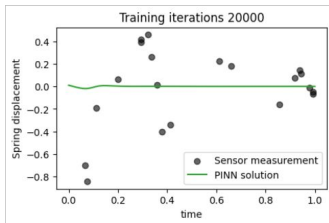
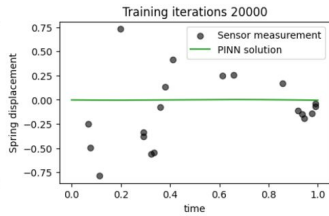
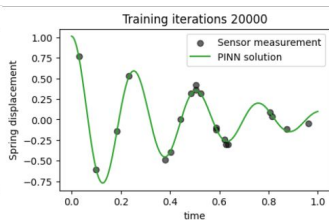
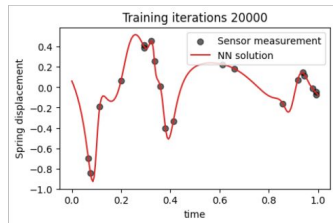
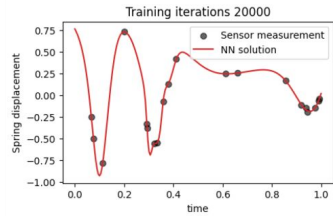
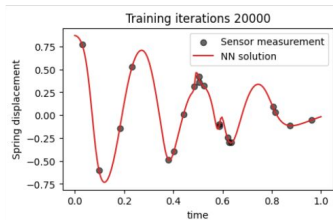
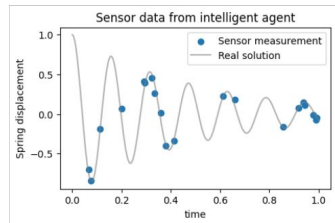
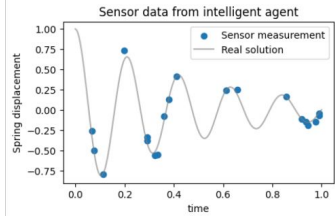
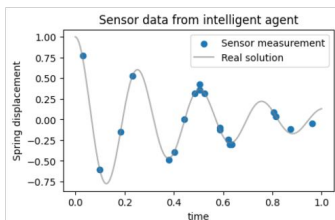


Anomaly 2



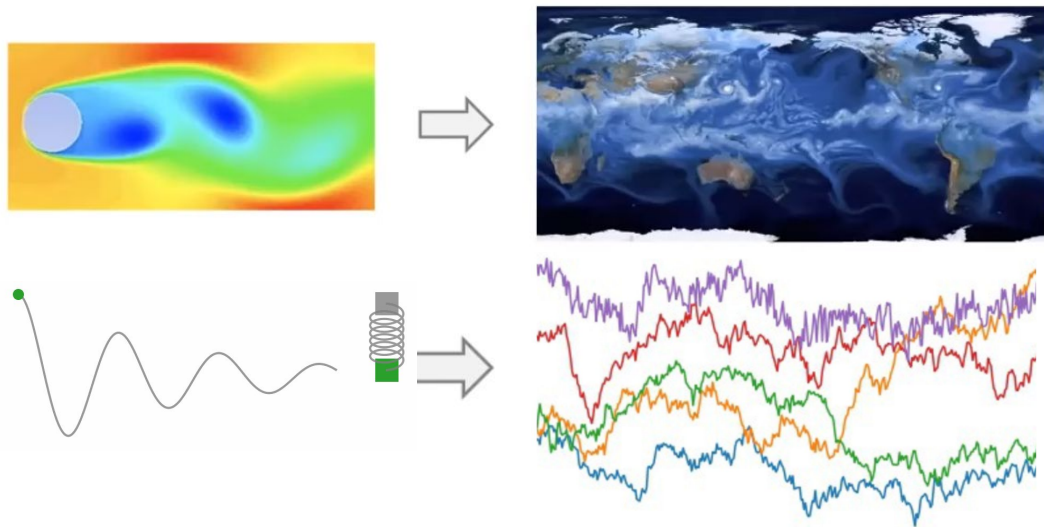
Hybrid AI for Anomaly Detection

Anomaly Detection - Additional Experiments



Hybrid AI for Anomaly Detection

PINNs - Limitations



Majority of PINN researches focus on toy or simplified problems

Difficulty scaling to:

- Large domains or higher frequencies (Spectral bias)
- Real world problems are much more complex (multi-physics)

Hybrid AI

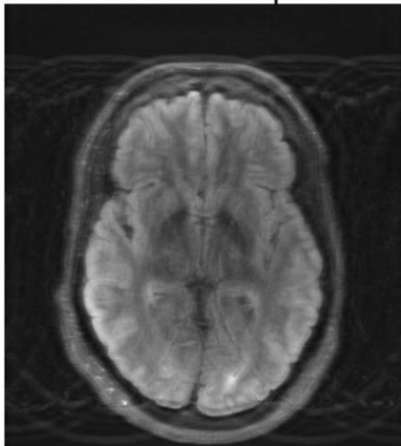
MIT researchers combine deep learning and physics to fix motion-corrupted MRI scans

The challenge involves more than just a blurry JPEG. Fixing motion artifacts in medical imaging requires a more sophisticated approach.

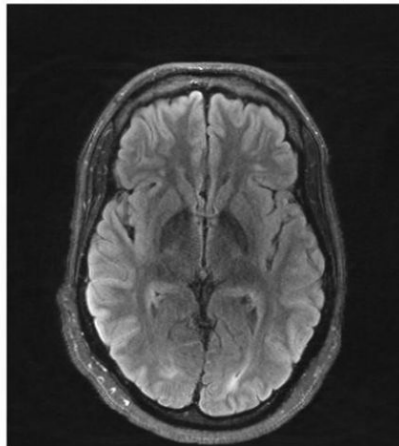
Alex Ouyang | Abdul Latif Jameel Clinic for Machine Learning in Health

August 17, 2023

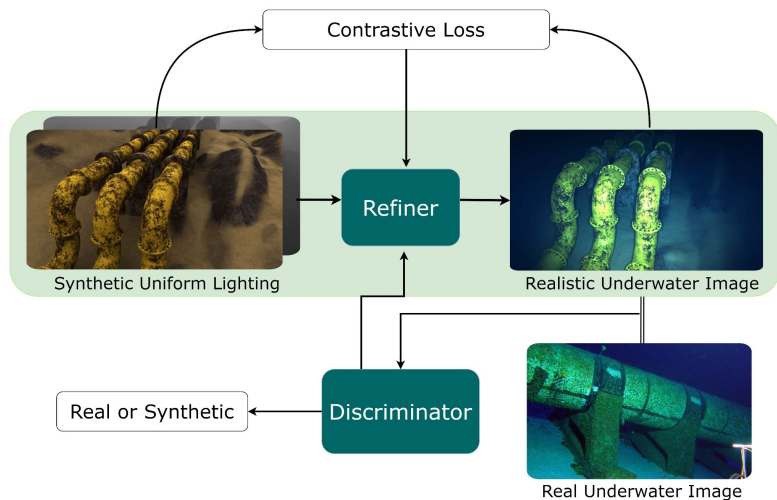
Motion-Corrupted



Motion-Corrected

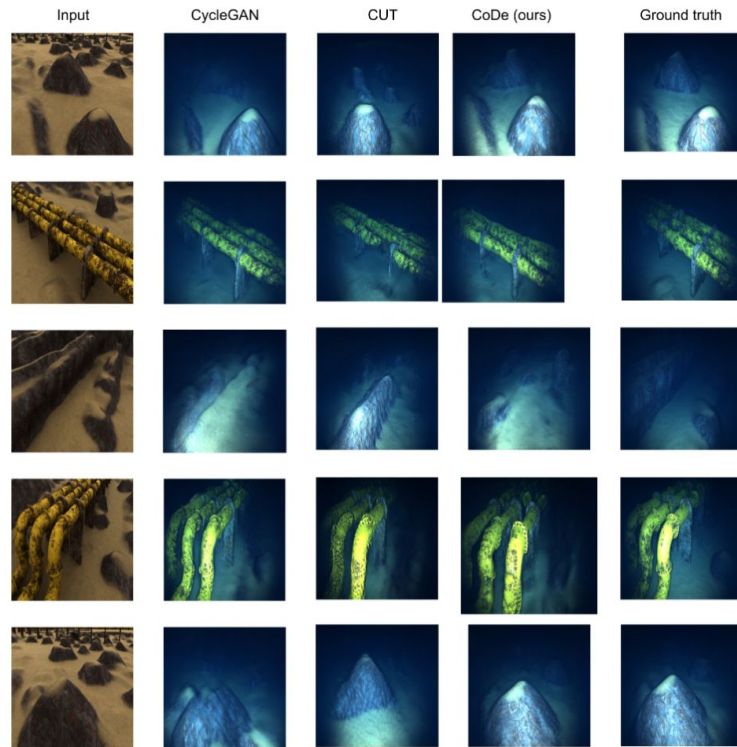


Hybrid AI



Using **GANs** to learn physical parameters from images

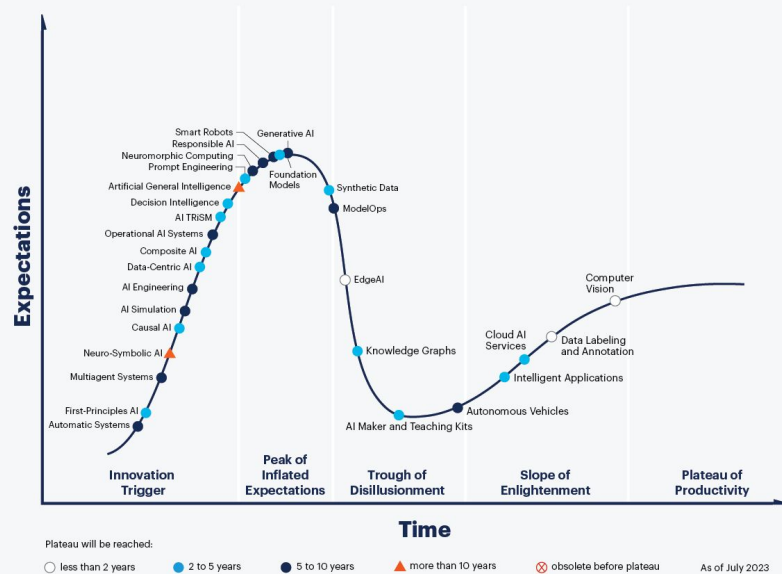
- Image Formation Model (IFM)
- Backscatter and Attenuation



Recap

- Anomaly detection techniques
- Imputation - Sensor Data Distribution
- Hybrid AI for Anomaly detection
- Scaling Hybrid to real world data
- Potentials and Limitations of Hybrid AI

Hype Cycle for Artificial Intelligence, 2023



Thank You!